NOTES FOR OPENTRON HACKATHON

TO-DO LIST:

Trisha, **Sofie**, **James**, **Salah**

- [ ] Fix app.py files so the output protocol files are correct (no duplications) (**Sofie**, **Trisha**)
- [x] ~~Fix GUI → Make prettier, include like a heading/title, TRY DATACLASS (**Sofie**, **Trisha**)~~
    - [x] ~~Fix dropdown so even though ot2 pipettes are pre.defined allow selection of other options as well~~
- [ ] Presentation (15 min max, What we want to do and why, our start position, errors we fixed, final outcome, GUI, universal file, pseudocode, demo video of clip reaction) **(Everyone)**
- [ ] Pseudocode (workflow of the clip reaction), How does our GUI strategy work **(Everyone)**
- [x] ~~Actual wet lab runs (Have to still run the Flex protocol) (**James, Salah**)~~
- [ ] Quantitative Analysis of our results (OT2 vs last weeks result) (**James, Salah**)
- [ ] Finish the GitHub Repo and post of the Opentrons work thingy
    - [ ] Finalise ReadMe (**Sofie**)
    - [ ] Make sure it runs in new environment (**Sofie**)
    - [x] ~~Add pseudocode to github~~
    - [x] ~~Add graphics and images of Ot2 and Flex setup~~
    - [x] ~~MAKE A LOGO FOR DNABOT~~
- [ ] SUBMIT GITHUB REPO, PSEUDOCODE, PRESENTATION (DEADLINE, DEC 10th 12PM!)

Slides:

1. **Intro into DNABOT**
    a. **What is the reason behind developing this app? What do we want it to do in the end?**
    b. **How this app will help with BASIC assembly in the long run**
2. **Schematic of how the app works**
3. **Ot-2 vs flex**
    a. **How are they different?**
    b. **What problems may this cause, and what are we trying to fix? (need different syntax - want to make a universal template file instead of separate files for flex and ot2)**
4. **Gui**
    a. **What it was before, and after - how was it optimized**
5. **app/template files (?)**
6. **Experiments**
    a. **Running the clip reactions on both ot-2 and flex**
7. **Volumetric tests**
8. **Final outcomes/summary slide**
9. **Advantages of our changes**
10. **Future works**

OPENTRON HACKATHON BASIC ASSEMBLY BACK-UP PLAN
Streamlined version of the original code/dnabot

Errors we ran into:

- Recognising API (2.0,2.20, 2.21, there were different versions within one file)
- Numpy can not be imported (deck slots are not correct, we apparently index plates by strings ⇒ OT-2 cannot do this)
- FLEX vs OT-2 (for the UI: when the user decides to run the code on OT-2 there should not be any dependencies in the code for FLEX

```
if robot_type=='Flex':
        trash = protocol.load_trash_bin("A3")
        #tc_mod = protocol.load_module(module_name=__HARDWARE['thermocycler']['id'], location = "B1")
        tiprack_type = ['opentrons_flex_96_tiprack_50ul']
        #tiprack_1000 = ['Flex_tiprack_1000ul'] 'opentrons_flex_96_tiprack_1000ul'
    elif robot_type=='OT-2':
        tiprack_type = __LABWARES['OT-2_tiprack_20ul']['id']
    Else: …
```

- Invalid pipette argument: push_out
- Protocol Name is set to FLEX

Ideas what to change/optimise:

- **Two distinct files between OT-2 and FLEX**
  - **Remove Hardware and Labware global registries**
  - **Minimise If statements**
  - **Use OT-2 or FLEX specific arguments**
- **Optimise UI**
  - **Have a final specific file that runs (no dependencies so when the user says its a 300 8 channel pipette this will just be copied into the template file so it is specific to the protocol you want to run…if that makes sense…)**
  - **Remove FELX options from Drop Down menu when the user chose OT-2 in the beginning**
- **Change Flow rates (change either in the Opentron app or add to code??)**

what was changed

- api error and changed

- the parts_vol was defined as parts_volume and had to be changes

- the source_vol was defined as source_volume and had to be changes

- mistake in the logic statement of the linker volume and part volume, safetz net for high amounts of whatever so when its above 100 or 40 we limit the amount we changed it so it actually runs

- there is a location type error for the bin

- there was a pipette error and was initially pipette_---

- from_centre_cartesian this piece of code will code the opentron for the flex and not OT-2 and so the .bottom will code for the OT-2 which the from_centre_cartesian had to be changed to .bottom

- now the file is stuck as a opentrons flex one and is now to be changed to a OT-2

- the if statements were corrected

- places an else statement in response to the code not working

- Protocol name was set to FLEX without any dependencies

- name of the file was changed

- Did not calibrate initially but will in future runs

-

Second protocol purification errors and changes
- Comma left on line 111, this caused an error to occur and was removed

- global function tip index error on line 309 (global tip index does not work)

- the tip index was removed from brackets in both flex and OT-2

- the api level was corrected in the metadata

- loading error was found for tip rack line 85 for local protocol

    The CANDIDATE_TIPRACK_100 was not defined for the OT_2 opentron

-error on line 89 for the tiprack 1000 this was undefined

- edits were made around line 72-83 and some comments were uncommented

- line 76 “96_tiprack_300ul” error was found protocol analysis error

- the tiprack_1000 is still not defined even in the metadata, was added in the labware, we then changed the code from then on where it spoke about the flex having 1000ul tips and also added it in for the OT-2

- like 117 max_volume - cannot attribute max_volume error

- pipette.max_volume 300 was commented out

-key error (line 123): ‘mag_plate’ - must be stated in the labware but it is not

- ‘mag_plate’[id] is placed in _labware, along with its 96 custom well plate

- value error in line [126]: modules are not recognised

- magneticmoduleV1 was placed in hardware metadata

- protocol failed command error line[128]: instead of protocol load labware it was changed to MAGDECK load labware

- type error [line 192]: unsupported operand

-sample_offset = int(sample_offset) was put into like 191

- now error on line [191] - the comma on sample_offset = 0, was removed on line [57]

- type error line [211] - cannot multiply

- elution buffer = 40 the comma was removed

- all commas will be removed after each integer value

- line 212 unsupported operand -

- commas were removed from line 45,47 and 48

- name error line [252] - ‘trash’ - removed trash from the brackets

- tip_index is not defined - replaced global with nonlocal

- attribute error [line 347] ‘list’ object has no attribute

- changed line 347 to have a [0] next to the word target

- indented lines 348-352

- unexpected indent line 359 - lines 356,357 and 359 were re indented

- line 361 is incorrectly indented and must be corrected

- from line 361 and onwards the lines were reindented

- protocol command failed error line 354 - problem with picking up the tip

- pick up tip line was added to line 340

- big change edit from line 350-355 - with the use of a script generated by Chat gpt

- attribute error [line 352] ‘list’ object has no attribute - must update the definition of ethanol

-all commas were deleted from unnecessary parts of code

- PROBLEM: THIS CODE MAKES THE OPENTRON TOO RUN SLOW (WE THOUGHT IT WAS BROKEN)

  - Likely due to slow() commands which don’t specify a speed; needs to be checked

https://github.com/BASIC-DNA-ASSEMBLY/DNA-BOT/blob/DNABOT-FLEX/dnabot/MRes2025/example/1_OT-2_clip_APIv2_21.py

Assembly protocol errors and changes:

1. Syntax errors such as unexpected indent in if statement block, commas after print statements
2. Same as clip reaction protocol, moved API to metadata directory and removed the requirements directory
3. TypeError [line 246]: run.<locals>.final_assembly() missing 2 required positional arguments: 'robot_type' and 'tiprack_type'
4. NameError [line 188]: name 'tube_rack' is not defined
    a. Defined tube_rack, destination_plate (variables used later in the code)

ERRORS FOR THE FLEX FILES:

1. CLIP FLEX
    a. LABWARES ARE WRONGLY DEFINED SO EVEN THOUGH ITS A FLEX PROTOCOL it still says OT-2 Pipettes and stuff
    b. wed

UI improvements:

1. Once you choose a robot type, default change to compatible labware/hardware
2. SCROLL WITH TRACKPAD/MOUSE NOT BAR
3.

salah & james 4/12/2025

- Made a custom labware part that is now on the imperial opentrons labware file, this custom labware part is a tube rack. This labware is now a custom labware on the opentrons app
- Edited and finalised the code for the clip reaction
- Placed PBS, flurocein in tubes to be used for the final clip reaction run on the opentron 2
- Made a dye solution using coumarin 30 (1mol, 1ul ethanol, 1ul PBS) , this solution was then diluted with another 1ul of ethanol
- Replaced the pipet tip on the opentron OT-2
- Ran clear run on the opentron, then ran water ran on opentron
-

04/12/25 - Trisha&Sofie

- Ran the GUI to output Flex-specific protocol files, and then edited it so that it is recognised as a Flex robot protocol on Opentrons.
- Then combined the edited code from the OT-2 Clip protocol file and the requirements directory to create a universal file (Attempted_UNI):

- The file ran on Opentrons, analyzed successfully and was recognised as a Flex protocol and not an OT2 protocol; something that was the problem previously.
    - It didn't previously as we were calling the Robot 'Flex' in the code but the actual robot was named 'Flextron xyx (something i cant remember) so it was not registering them both as the same thing- so defaulted to thinking the code was for OT-2. ***FIXED***: renamed the robot 'Flex' on the actual machine.
- This created an error where the GUI's output files only listed OT-2 specific labware - so if robot_type–'Flex' then it would still call from the metadata and __LABWARES dictionaries which only specify OT-2 labwares.
    - FIX:

- When loading the app, it required the installation of many packages and environments separately. To fix this, we put all the packages that need to be installed into one 'requirements' file that can be called to install when running the dnabot_app2_1.py file.

Salah & james - 05/12/2025

- Ran a full liquid run using the coumarin 30 as DNA the linkers as flurocein and both the master mix and the water were PBS
- One ran a gradient was made initially using the c30 going up from 0-10 and the F going down from 10-0 with PBS on either side
- The plates were then fluorescent scanned, this had shown that the c30 was way too big of a concentration so the c30 has to be diluted
- We then diluted the c30 by 10 fold having 100ul of the c30 with 900ul of ethanol
- The opentron was ran again using this concentration
- The gradient was then made on the PCR plates with the layout shown below
- The plates were then scanned and had shown that the gradient was reasonable and this shows that it was efficient
- We also calibrated the flex machine
- Replaced the 1000ul single pipette with a 50ul single pipette on the flex machine

What to do next
Code a gradient for the opentron to do inorder to test the accuracy
What is what
linkers - fluorescent dye absorption 494 and em 510-520
Master Mix - PBS
DNA - Coumarin 30 absorbance 400 emission 480
Overlap may occur
Must have control on the side before fluorescent scanning the plate
The 1m of C30 was mixed with 1ml of pbs and 1ml of ethanol
Was then mixed and centrifuged
0.25-0.3ml of the supernatant was then mixed with 1ml of ethanol

Control

	1	2	3	4	5	6	7	8	9	10	11	12
A	T, 2.5	T, 1.25	PBS	0C-2F	0.1C-2F	0.2C-2F	0.5C-2F	1C-2F	2C-2F	2.5C-2F	0C-0.5F	0C-1F
B	T, 1	T, 1	1C-0F	1C-0.5F	1C-1F	1C-1.5F	1C-2F	1C-2.5F	-	-	-	-
C	T, 1	T, 1.25	-	-	-	-	-	-	-	-	-	-
D	T, 1	T, 1	-	-	-	-	-	-	-	-	-	-
E	T, 1.25	-	-	-	-	-	-	-	-	-	-	-
F	T, 1.25	-	-	-	-	-	-	-	-	-	-	-
G	T, 1	-	-	-	-	-	-	-	-	-	-	-
H	T, 1	-	-	-	-	-	-	-	-	-	-	-

DNA parts (coumarin 30)
Linkers (fluorescein)
Each well has 2ul fluorescein
Each well has 30ul of PBS

	1	2	3	4	5	6	7	8	9	10	11	12
A	PBS	0C-2 F	1C-2 F	2C-2 F	3C-2 F	4C-2 F	5C-2 F	6C-2 F	-	-	-	-
B	-	-	-	-	-	-	-	-	-	-	-	-
C	-	-	-	-	-	-	-	-	-	-	-	-
D	-	-	-	-	-	-	-	-	-	-	-	-
E	-	-	-	-	-	-	-	-	-	-	-	-
F	-	-	-	-	-	-	-	-	-	-	-	-
G	-	-	-	-	-	-	-	-	-	-	-	-
H	-	-	-	-	-	-	-	-	-	-	-	-

Fluorescence scan shows value goes up with the increase in coumarine30
Need to come up with a better procedure.

Add back in line 82-106 for the thermocycler off code

HOW TO GET THE REPO ON LOCAL VS CODE:

py --version (or if that does not work python --version
where pip (to see if we have pip)
Download python 3.10 from website or use homebrew for macos

Create an environment:
python3.10 -m venv venv
.\venv\Scripts\activate

Pip install -r requirements.txt

On Macos:
Brew install python@3.10
Brew link python@3.10 --force
Python3.10 --version

Python3.10 -m venv venv
Source venv\bin\activate
Pip install -r requirements.txt

To see which python version is running:
py --list
py -3.10

py -3.10 -m venv venv

Notes for the GUI:

- **WHen selecting CSV files you cannot see which files were selected**
- **Gret background is boring**
- **Modernise it more**
- **Add a title thingy to the top of the page**
- **Default is correct but we still need to be able to select other options so fo agar plate it just shows the one**